

RIDHI SRIKANTH

[LinkedIn](#) | [GitHub](#) | [Web Portfolio](#) | Phone :+1 (425) 922-4763 | ridhisrikantth23@gmail.com

SUMMARY

Curiosity drives me to explore, and determination ensures I follow through. I am a Computer Engineering student at UCSD, engaged in software development and CS research. I learn quickly and turn ideas into results, persisting through challenges to achieve success.

EDUCATION

University of California, San Diego - BS, Computer Engineering **Graduation:** June 2027 GPA: 3.99, Provost Honors
Selected Coursework: Advanced DSA, Operating Systems, Parallel Programming, Computer Architecture, Machine learning

AWARDS AND HONORS

Eta Kappa Nu (HKN) – IEEE Honor Society - Inducted for academic excellence in Engineering
Winner of UCSD WIC programming competition Fall 2023 - won 1st place among 100+ participants for accuracy and speed

SKILLS

- **Programming Languages:** Java, Python, C, C++, OpenCL, MySQL, TypeScript, JavaScript, HTML, SQL, CSS, Shell, Rust
- **Technologies and Expertise:** React, Flask, Django, Agile, CI/CD, Git, Linux, Embedded Systems, Firebase

LEADERSHIP

- IEEE UCSD Chapter Technical Chair: Led 5+ technical workshops, including a live-coded React web dev session and a recommendation systems workshop, fostering hands-on learning for peers.

EXPERIENCE

Deep Learning for Food Recognition - Internship Health-tech Startup (Stealth) July 2025 - Aug 2025

- Deployed a PyTorch CNN for food image recognition and nutrient computation, reducing Gemini token usage by 50%.
- Trained the in-house model on user-submitted images and datasets, with expert verified labels to ensure accuracy.
- Images not recognized by Convolutional Neural Network (CNN) are routed to Gemini-2.5-Flash.

Research Assistant - UC San Diego CSE Department (Professor Michael Coblenz)

Project 1: Scientific Workflow Reproducibility Study Feb 2025 - July 2025

- Collaborated with a research team on a corpus study of 500+ scientific programs; Analyzed publicly available code bases for runtime errors, debugged issues and simulated the code to study software reproducibility.
- Co-authored a research paper

Project 2: Enhancing Shell Scripting Workflows by Modifying Python Compiler in C June 2025 - Present

- Leading an independent research project; performed thematic and static analysis of shell scripts from scientific workflows
- Leveraging findings to design a Python-based Domain Specific Language (DSL) for safer and accessible shell scripting
- Working across CPython's compiler pipeline: modifying PEG grammar, parser logic in C, AST generation, bytecode emission, and stack behavior to integrate new language features.
- Writing C functions and updating VM evaluation logic to implement the DSL's execution semantics.
- Using GDB and backward-tracing to debug low-level issues, including stack inconsistencies and buffer overflows.

ECE 35 (Analog Design) Course Tutor Sep 2025 - Present

- Managing support for 400+ students, providing tutoring and hands-on lab guidance on real device circuit implementation
- Collaborating in weekly meetings to refine instructional strategies and responsible for grading quizzes and final exam questions of fundamental circuit theory concepts.

Full-stack Developer - ElectroTriton With Professor Curt Schurgers | [ElectroTriton](#) Aug 2025 - Present

- Developing features in ElectroTriton, a Django and MySQL homework management web app used for Engineering courses
- Implemented course cloning functionality which accomplished a ~71% reduction in course setup time (35→10 min)
- Automated student list updates which reduced roster management time by ~78% (45 → 10 min).

Full-stack Developer - LabStream With Professor Saharnaz Baghdadchi | [LabStream](#) Oct 2024 - June 2025

- Developed, with team of 8, a scalable web application enabling real-time remote access to electrical engineering labs
- Built a real-time slot booking system that reduced scheduling time for lab sessions by ~65% (multiple emails → 5 mins)
- Implemented features including lab session creation, active user session management, and role-based access control.
- Evaluated and integrated Firebase for data storage, CRUD operations, authentication; developed reusable React components

PROJECTS

Hardware Developer | [Hardware Blog](#): Designed and developed 5+ embedded systems with C++ firmware, custom circuit design, soldered prototypes and PCB layouts; blog followed by 2,000+ enthusiasts.

Game Developer | [Portfolio](#): Built 4 video games ranging from 3D platformers to 2D shooters and participated in 2 game jams.